

# MMF-1 Exam Preparation Guide / Подготовка к экзамену ММФ-1

Mustakim

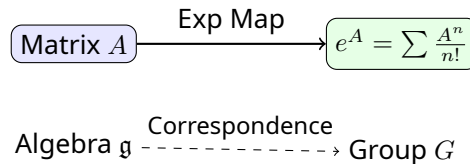
## Содержание

|          |                                                           |          |
|----------|-----------------------------------------------------------|----------|
| <b>1</b> | <b>Finite Groups (Конечные группы)</b>                    | <b>2</b> |
| 1.1      | 1. Matrix Function, Matrix Exponent . . . . .             | 2        |
| 1.2      | 2. Group Definitions (Groups, Subgroups, Order) . . . . . | 2        |
| 1.3      | 3. Group $D_3$ (Triangle Symmetry) . . . . .              | 2        |
| 1.4      | 4. Isomorphism and Homomorphism . . . . .                 | 2        |
| 1.5      | 5. Representations and Characters . . . . .               | 2        |
| 1.6      | 6. Irreps of $C_n, D_3$ . . . . .                         | 3        |
| 1.7      | 7. Decomposition into Irreps . . . . .                    | 3        |
| 1.8      | 8. Molecular Vibrations . . . . .                         | 3        |
| <b>2</b> | <b>Continuous Groups (Непрерывные группы)</b>             | <b>3</b> |
| 2.1      | 1. Lie Group Definition, Generators . . . . .             | 3        |
| 2.2      | 2. Generators on Smooth Functions . . . . .               | 3        |
| 2.3      | 3. Dimensions of Classical Groups . . . . .               | 4        |
| 2.4      | 4. $SO(2), SO(3)$ . . . . .                               | 4        |
| 2.5      | 5. Clebsch-Gordan . . . . .                               | 4        |
| <b>3</b> | <b>Analytic DEs (Аналитическая теория ДУ)</b>             | <b>4</b> |
| 3.1      | 1. Singular Points (Classification) . . . . .             | 4        |
| 3.2      | 2. Asymptotics . . . . .                                  | 5        |
| 3.3      | 3. Separation of Variables . . . . .                      | 5        |

# 1 Finite Groups (Конечные группы)

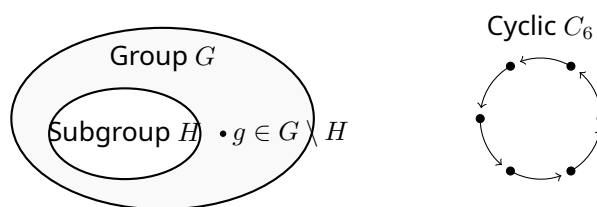
## 1.1 1. Matrix Function, Matrix Exponent

**Def:** Let  $f(z) = \sum c_k z^k$ . For matrix  $A$ :  $f(A) = \sum_{k=0}^{\infty} c_k A^k$ . **Matrix Exponent:**  $e^A = I + A + \frac{A^2}{2!} + \dots$  **Properties:**  
 1.  $[A, B] = 0 \implies e^{A+B} = e^A e^B$ . 2.  $\det(e^A) = e^{\text{Tr}(A)}$ .



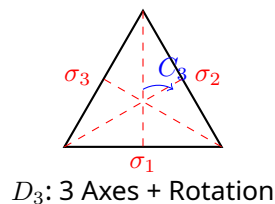
## 1.2 2. Group Definitions (Groups, Subgroups, Order)

**Group:** Set  $G$  with operation  $\cdot$  (Closure, Assoc, Identity, Inverse). **Subgroup:**  $H \subseteq G$ , closed under operation. **Cyclic:** Generated by  $a$  ( $G = \{e, a, a^2, \dots\}$ ).



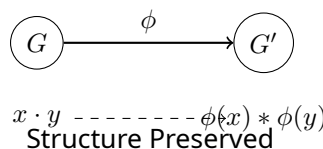
## 1.3 3. Group $D_3$ (Triangle Symmetry)

$D_3 \cong S_3$ . Order 6. Elements:  $E, 2C_3, 3\sigma_v$ . **Invariant Subgroup:**  $C_3$  (Rotations). **Cosets:** Partition  $G = H \cup gH$ .



## 1.4 4. Isomorphism and Homomorphism

**Homomorphism:**  $\phi(xy) = \phi(x)\phi(y)$ . **Isomorphism:** Bijective homomorphism.



## 1.5 5. Representations and Characters

**Rep  $D(g)$ :** Map  $G \rightarrow GL(V)$ . **Character:**  $\chi(g) = \text{Tr}(D(g))$ . Constant on classes.

$$\text{Element } g \xrightarrow{D(g)} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\chi(g) = a + d$$

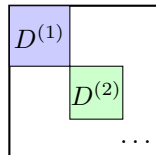
## 1.6 6. Irreps of $C_n, D_3$

$C_n$ : 1D reps  $\chi_k(m) = e^{2\pi i k m / n}$ .  $D_3$ :  $1^2 + 1^2 + 2^2 = 6$ . Two 1D ( $A_1, A_2$ ), one 2D ( $E$ ).

## 1.7 7. Decomposition into Irreps

Representation  $D$  reduces to block-diagonal form.

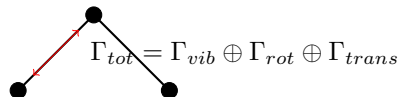
$$D \cong \bigoplus n_i D^{(i)}, \quad n_i = \frac{1}{|G|} \sum \chi^{(i)*} \chi$$



Block Diagonal Form

## 1.8 8. Molecular Vibrations

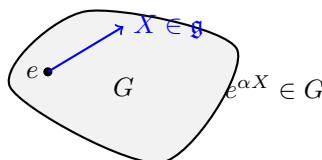
$$\chi_{vib} = N_{fixed} \cdot \chi_{vec} - \chi_{trans} - \chi_{rot}.$$



# 2 Continuous Groups (Непрерывные группы)

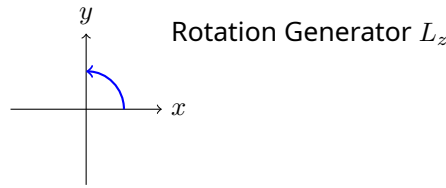
## 2.1 1. Lie Group Definition, Generators

Group + Manifold.  $g(\alpha) \approx I + \alpha X$ . **Generator  $X$** : Tangent vector at identity.



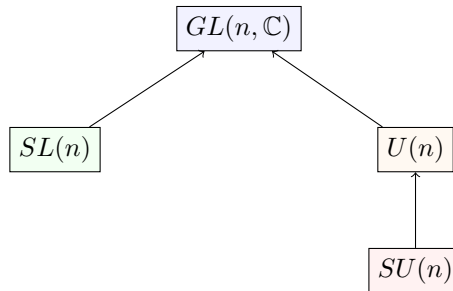
## 2.2 2. Generators on Smooth Functions

$$L\psi = \frac{d}{d\phi} \psi(g^{-1}x). \text{ For } SO(3): \text{Rotation around axis} \implies L_z = -i(x\partial_y - y\partial_x).$$



### 2.3 3. Dimensions of Classical Groups

$GL(2n^2)$ ,  $SL(2n^2 - 2)$ ,  $U(n^2)$ ,  $SU(n^2 - 1)$ ,  $SO(n(n - 1)/2)$ .

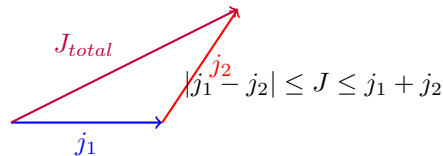


### 2.4 4. SO(2), SO(3)

$SO(2)$ : Circle  $S^1$ , abelian.  $SO(3)$ : Ball with identified surface points. Irreps  $SO(3)$ :  $D^{(l)}$ ,  $\dim 2l + 1$ .

### 2.5 5. Clebsch-Gordan

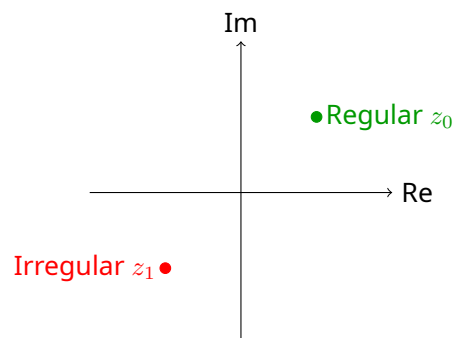
$D^{(j_1)} \otimes D^{(j_2)} \rightarrow \oplus D^{(J)}$ . Triangle rule  $|j_1 - j_2| \leq J \leq j_1 + j_2$ .



## 3 Analytic DEs (Аналитическая теория ДУ)

### 3.1 1. Singular Points (Classification)

$w'' + p(z)w' + q(z)w = 0$ . Regular Singular: Pole order  $p \leq 1, q \leq 2$ .



Fuchs Condition:  $\lim_{z \rightarrow z_0} (z - z_0)p < \infty$

## 3.2 2. Asymptotics

Regular:  $w \sim (z - z_0)^\rho$ .  $\rho$  from indicial eq. Irregular:  $w \sim e^{1/z}$  (Essential singularity).

## 3.3 3. Separation of Variables

$\psi(x, y, z) = X(x)Y(y)Z(z)$ . Used for PDE  $\rightarrow$  ODE.

